MTH1004

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2021

Probability, Statistics and Data

Module Leader: Prof. Daniel B. Williamson

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination.

SECTION A

1. (a) There are 60 lorries queueing at Dover attempting to deliver goods to the EU. The UK-EU trade deal requires specific paperwork for different types of goods. The main ones are "Excise duty", "Phytosanitary Certification" and "Export Health Certification". All of the lorries in the queue carry goods that require at least one of these types of paperwork at the border. Of the lorries, 36 carry goods requiring export health certification, 27 carry goods requiring phytosanitary certificates and 24 carry goods requiring excise duty. Additionally, 15 require both phytosanitary and export health certification, 6 require excise duty and phytosanitary certification, and 6 require excise duty and export health certification. If picking a lorry at random from the queue, what is the probability that they carry goods requiring all 3 types of paperwork? (10)

   (b) Let A and B be events with probabilities $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$. Find the smallest interval $[a, b]$ such that $P(A \cup B) \in [a, b]$. (11)

   (c) Two discrete random variables X and Y, both with range spaces $\{1, 2, 3\}$ have the following joint probability mass function (pmf)

$p_{X,Y}(x, y)$	$x = 1$	$x = 2$	$x = 3$
$y = 1$	c	.10	.00
$y = 2$	.10	c	.10
$y = 3$	.00	.10	c

   where c is a constant.

   (i) Calculate c.

   (ii) Calculate the marginal pmf $p_X(x)$ and the marginal expectation $E[X]$.

   (iii) For each $y \in \{1, 2, 3\}$, calculate the conditional pmf of X given $Y = y$.

   (iv) Calculate the conditional expectation of X given $Y = 2$. (12)

(d) To investigate the effect on athletic performance of a new training regime, the times taken by each of 30 athletes to run 400 metres were recorded before and after adopting the new regime. The average change in time of the 30 athletes was $\bar{y} = -0.20$ seconds and the standard deviation of the changes was $s = 0.42$ seconds.

(i) Conduct a test of size 1% of the null hypothesis that there is no change in the population mean time against the alternative hypothesis that the population mean time reduces. State clearly your test statistic, its null distribution, the critical region and your conclusion about the effect of the new training regime.

(ii) Compute the p-value for this test.

(iii) Compute an equal-tailed 99% confidence interval for the population mean change in running time. (17)

[**50**]

SECTION B

2. Let X_1 and X_2 be independent, continuous Unif$(0, 1)$ random variables and let $Y_1 = X_1 + X_2$. Calculate the probability density function of Y_1. **[25]**

3. (a) Consider an experiment in which the outcomes of control and intervention groups are compared. In this context, explain briefly what is meant by 'confounding', 'matching' and 'randomizing', and how matching and randomizing reduce the chance of confounding. (12)

   (b) To investigate the effect of a feed supplement on the weights of horses, the supplement will be added to the diet of n horses and will be omitted from the diet of another n horses. The change in weight after one month will be measured for each horse. A test will be conducted to determine whether the supplement produces heavier horses. Let μ_1 and μ_2 denote the population mean changes in weight for horses fed with and without the supplement, respectively. Assume that n will be large and that the standard deviation of weight changes for both groups of horses is 20 kilograms.

      (i) Write down a suitable test statistic and its null distribution, defining any notation that you use.

      (ii) The size of the test will be 1%. Derive an expression for the power of the test and hence compute the sample size, n, that would be required for the power to be at least 80% when $\mu_1 - \mu_2 = 10$ kilograms. (13)

   [25]

4. During lockdown Gary is invited to a friend's house to make up an 8 player poker game. To understand the risks of exposure to Covid-19, he finds out that Exeter has 300 cases per 100000 people. He has heard that 1/3 of Covid-19 infections are asymptomatic, and guesses that those people wouldn't know to get tests and so wouldn't be counted in those figures, so it is probably more likely to be 450 cases per 100000 people. Gary reasons that the probability that a random person in Exeter (and therefore at the poker game) has Covid-19 is 0.0045.

   (a) Using Gary's numbers, what is the probability that none of the other players at the game has Covid-19? (3)

   (b) Deciding not to break the law, Gary decides to go for a run instead. He knows that "on-average" he will run past 10 people an hour, so models the number, X, of people he would run past per hour as Poisson with rate $\lambda = 10$. Let Y be the number of people he runs past in an hour who would test positive for Covid-19 so that $Y \mid X \sim \text{Bin}(X, p)$ where $p = 0.0045$ (for Gary). Show that the joint probability mass function of X and Y is

$$p_{XY}(x, y) = P(Y = y, X = x) = \frac{\lambda^x e^{-\lambda}}{y!(x-y)!} p^y (1-p)^{x-y}.$$

   (6)

   (c) Hence show that $Y \sim \text{Pois}(p\lambda)$. (*Hint: you may use the fact that* $e^\theta = \sum_{j=0}^{\infty} \frac{\theta^j}{j!}$). (8)

   (d) Write down an appropriate model for the time until Gary passes a Covid-19 infected individual when out running. (3)

   (e) What is the expected time he could be out before passing someone with Covid-19? (2)

   (f) If he is out for 1 hour and 45 minutes, what is the probability that he passes no one with Covid-19? (3)

   [**25**]